

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 2

Year 12 Higher School Certificate
Trial Examination Term 3 2023

STUDENT NUMBER: _____

General Instructions

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
Black pen is preferred
- NESA-approved calculators and drawing templates may be used
- A reference sheet is provided separately
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination room

Total marks – 100

Section I Pages 3 – 6

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet
provided

Section II Pages 7 – 12

90 marks

Attempt Questions 11 – 16

Start each question in a new writing booklet

Write your student number on every writing booklet

Question	1-10	11	12	13	14	15	16	Total
Total	/10	/15	/15	/15	/15	/15	/15	/100

This assessment task constitutes 30% of the Higher School Certificate Course School Assessment

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

1 The scalar product of two vectors $\underline{u} = 3\underline{i} - \underline{j} - 7\underline{k}$ and $\underline{v} = 2\underline{i} + 3\underline{j} + \underline{k}$ is

(A) $6\underline{i} - 3\underline{j} - 7\underline{k}$

(B) $5\underline{i} + 2\underline{j} - 6\underline{k}$

(C) -4

(D) 22

2 If $z = 3 - 4i$, then $\frac{1}{1-z}$ is equal to:

(A) $\frac{-1-2i}{10}$

(B) $\frac{-1+2i}{10}$

(C) $\frac{-1-i}{6}$

(D) $\frac{-1+i}{6}$

3 The equation of a sphere, centre $(2, 2, 3)$ that touches the $x-y$ plane only is:

(A) $(x-2)^2 + (y-2)^2 + (z-3)^2 = 9$

(B) $(x-3)^2 + (y-2)^2 + (z-2)^2 = 9$

(C) $(x-2)^2 + (y-2)^2 + (z-3)^2 = 6$

(D) $(x-2)^2 + (y-3)^2 + (z-3)^2 = 9$

4 Which of the following is equivalent to the expression $\frac{16x-43}{(x-3)(x^2-x-6)}$?

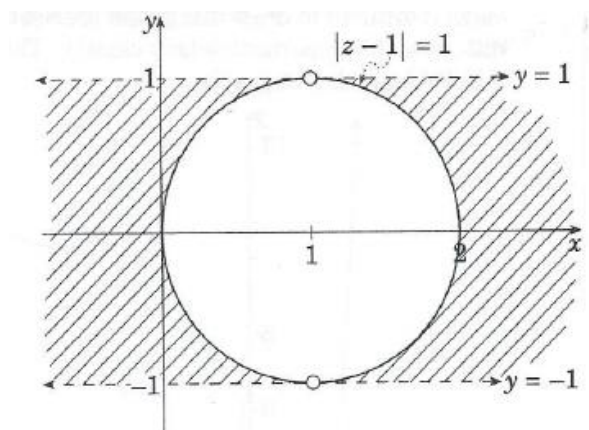
(A) $\frac{3}{x-3} + \frac{1}{x-6} - \frac{3}{x+1}$

(B) $\frac{3}{x-3} + \frac{1}{(x-3)^2} - \frac{3}{x+2}$

(C) $\frac{3}{x-3} + \frac{1}{x+3} - \frac{3}{x-2}$

(D) $\frac{3}{x-3} + \frac{1}{x+2} - \frac{3}{(x+2)^2}$

5 Which of the inequalities best represents the given region in the complex plane?



NOT TO
SCALE

(A) $\{z : |z + \bar{z}| < 2\} \cup \{z : |z - 1| \geq 1\}$

(B) $\{z : |z + \bar{z}| < 2\} \cap \{z : |z - 1| \geq 1\}$

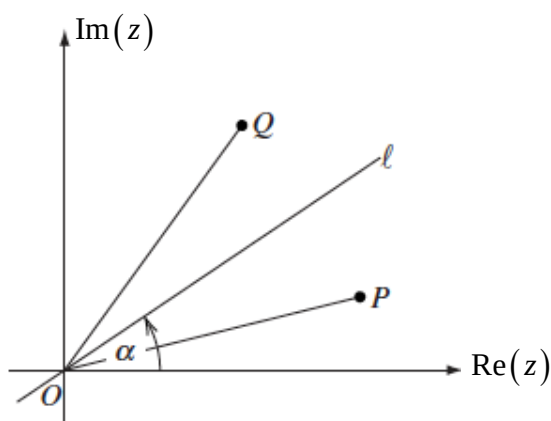
(C) $\{z : |z - \bar{z}| < 2\} \cup \{z : |z - 1| \geq 1\}$

(D) $\{z : |z - \bar{z}| < 2\} \cap \{z : |z - 1| \geq 1\}$

6 Which expression is equivalent to $\int (\ln x)^2 dx$

- (A) $x(\ln x)^2 - 2 \int \ln x dx$
- (B) $(\ln x)^2 - 2 \int \ln x dx$
- (C) $x(\ln x)^2 - 2 \int x \ln x dx$
- (D) $2 \int \ln x dx$

7 l is a line in the Argand diagram that passes through the origin and makes an angle α with the positive real axis, where $0 \leq \alpha \leq \frac{\pi}{2}$.



The point P represents the complex number z_1 , where $0 < \arg(z_1) < \alpha$. The point P is reflected in the line l to produce the point Q , which represents the complex number z_2 .

Hence $|z_1| = |z_2|$.

The value of $\arg(z_1) + \arg(z_2)$ is

- (A) α
- (B) 2α
- (C) 3α
- (D) none of the above.

8 What is the converse of the following statement?

“If both m and n are divisible by d , then $m - n$ is divisible by d .”

- (A) If $m - n$ is divisible by d , then both m and n are divisible by d .
- (B) If $m - n$ is not divisible by d , then both m and n are not divisible by d .
- (C) If $m - n$ is not divisible by d , then neither m nor n is not divisible by d .
- (D) If both m and n are not divisible by d , then $m - n$ is not divisible by d .

9 Consider the statement:

‘ $\forall x \in R$, if $f(x)$ is even, then $f''(x)$ is also even’

Which of the following is the negation to the statement?

- (A) $\exists x \in R$, if $f(x)$ is not even, then $f''(x)$ is also not even
- (B) $\exists x \in R$, if $f''(x)$ is not even, then $f(x)$ is also not even
- (C) $\forall x \in R$, if $f(x)$ is not even, then $f''(x)$ is also not even
- (D) $\forall x \in R$, if $f''(x)$ is not even, then $f(x)$ is also not even

10 Which best describes the following parametric equations?

$$x = -5 \cos t$$

$$y = 5 \sin t$$

$$z = t$$

- (A) A spiral around the y axis moving in an anticlockwise direction.
- (B) A spiral around the y axis moving in a clockwise direction.
- (C) A spiral around the z axis moving in an anticlockwise direction.
- (D) A spiral around the z axis moving in a clockwise direction.

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet

(a) (i) Given that $a > 0, b > 0$, prove that $a + b \geq 2\sqrt{ab}$ 2

(ii) Hence show $\sec^2 x \geq 2 \tan x$ 1

(b) (i) Express $z = \sqrt{2} - i\sqrt{2}$ in modulus-argument form. 2

(ii) Hence evaluate z^{24} 1

(c) The point P represents the complex number $-2 + 4i$. The point Q is in the first quadrant of the Argand diagram such that $\angle OQP = \frac{\pi}{2}$ and $|OQ| = |QP|$. 2

Find the complex number represented by the point Q .

(d) Find the primitive function of each of the following:

(i) $\int \frac{e^x}{1+e^{2x}} dx$ 2

(ii) $\int \sin^2 x \cdot \cos^3 x dx$ 2

(iii) $\int \frac{2x+1}{x^2+2x+2} dx$ 3

End of the Question 11.

Question 12 (15 marks) Start a new writing booklet

(a) (i) Show that one of the zeros of the equation $2z^3 - 5z^2 + 6z - 2 = 0$ is $1 - i$. 1

(ii) Hence, or otherwise, find the other zeros over the complex field $\mathbb{C}$. 2

(b) Consider the statement 'If $5^x = 2$, $x \in \mathbb{R}$, then x is irrational.'

(i) Find the converse of the above statement. 1

(ii) Prove that the above statement is true. 2

(c) Find $\int x \sin^{-1} x \, dx$ 4

(d) (i) Point P divides the line segment AB internally in the ratio of $m:n$, prove that 2

$$\underline{p} = \frac{m\underline{b} + n\underline{a}}{m + n}$$

where $\overrightarrow{OP} = \underline{p}$, $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$.

(ii) Point P is the midpoint of AB , show that $\underline{p} = \frac{\underline{a} + \underline{b}}{2}$. 1

(iii) Points P , Q and R are midpoints of the sides of $\triangle ACB$, where $\overrightarrow{OC} = \underline{c}$. 2

Show that $\underline{p} + \underline{q} + \underline{r} = \underline{a} + \underline{b} + \underline{c}$.

End of Question 12

Question 13 (15 marks) Start a new writing booklet

(a) The line l_1 passes through the points $A(4, -3, -3)$ and $B(5, 2, 2)$.

(i) Find the vector equation of line l_1 . 2

(ii) The equation of line l_2 is given by $\frac{x-1}{k} = \frac{y}{-1} = \frac{z+3}{1}$ 3

Find the value of k for which l_1 and l_2 intersect.

(b) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{3 - \cos x - 2 \sin x}$ 4

(c) The spheres $x^2 + y^2 + z^2 = 9$ and $x^2 + (y-4)^2 + z^2 = 16$ intersect. Find

(i) the value of y when they intersect. 1

(ii) the equation of the circle in which they intersect, giving the coordinates of the centre and the radius. 2

(d) Use mathematical induction to prove for all integers $n \geq 2$, 3

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$$

You may assume $\frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0$ is true.

End of Question 13

Question 14 (15 marks) Start a new writing booklet

- (a) Use a counter-example to show that the statement ‘a quadrilateral is formed by joining any four points in a plane.’ 1
- (b) ‘If $\forall x \in R, x^2 - 6x + 5$ is even, then x is odd.’
- (i) Write down a contrapositive statement. 1
- (ii) Use contraposition to prove the original conjecture. 2
- (c) (i) Sketch the set of points representing the complex number z such that 2
- $$\arg(z + 2) = \frac{\pi}{4} + \arg(z - 2i)$$
- (ii) Find the radius and centre. 2
- (d) (i) If ω is a complex root of $z^3 - 1 = 0$ with the smallest positive argument, show that ω^2 is also a cube root of unity. 2
- (ii) Show that $1 + \omega + \omega^2 = 0$. 1
- (iii) Find all the roots in the form $e^{i\theta}$ and indicate these roots in an Argand diagram. 2
- (iv) Show that $(1 + \omega^2)^5 = -\omega^2$ 2

End of Question 14

Question 15 (15 marks) Start a new writing booklet

(a) Given $a^2 + b^2 \geq 2ab$ (Do NOT prove this)

(i) Prove that $x^2 + y^2 + z^2 \geq xy + yz + zx$ **1**

(ii) Deduce that if $x + y + z = 1$, then $xy + yz + zx \leq \frac{1}{3}$ **2**

(b) (i) Given $z = \cos \theta + i \sin \theta$, prove $z^n + \frac{1}{z^n} = 2 \cos n\theta$ **2**

(ii) Hence by considering the expansion $\left(z + \frac{1}{z}\right)^4$ show that **3**

$$\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$$

(iii) Hence find the primitive of $\int \cos^4 \theta \, d\theta$ **1**

(c) Find the shortest distance from the point $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ to the line $r = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ **3**

(d) A sequence is defined by $a_1 = 2$, $a_2 = 3$ and $a_n = 3a_{n-1} - 2a_{n-2}$ for all integers $n \geq 3$ **3**

Prove by mathematical induction that $a_n = 2^{n-1} + 1$ for integers $n \geq 1$

End of Question 15

Question 16 (15 marks) Start a new writing booklet

(a) Find the integral $\int_0^{\frac{\pi}{4}} \sec^3 x \, dx$. 3

(b) Lines l_1 and l_2 are given below, relative to a fixed point O .

$$l_1 : r = (11\hat{i} + 2\hat{j} + 17\hat{k}) + \lambda(-2\hat{i} + \hat{j} - p\hat{k})$$

$$l_2 : r = (-5\hat{i} + 11\hat{j} + q\hat{k}) + \mu(-3\hat{i} + 2\hat{j} + 2\hat{k})$$

where λ and μ are scalar parameters.

(i) Given that lines l_1 and l_2 are perpendicular, find the value of p . 2

(ii) Given that lines l_1 and l_2 intersect perpendicularly, find the value of q . 3

(iii) Hence find the point of intersection of lines l_1 and l_2 . 1

(c) (i) Evaluate $\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} \, dx$ 2

(ii) Show that $0 \leq x^n \leq x^2$, for $n \geq 2$ in the above integral. 1

(iii) Hence show that for $n \geq 2$, $\frac{1}{2} \leq \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^n}} \leq \frac{\pi}{6}$. 3

End of Examination

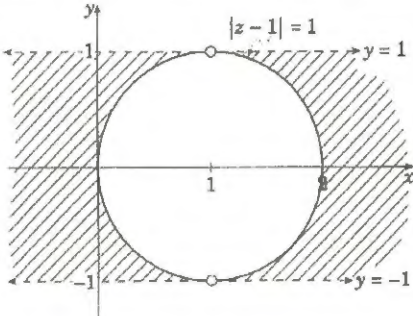
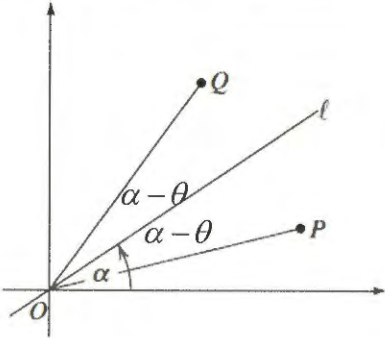
HGHS Ext 2 2023 TRIAL HSC SOLUTIONS

Marking Guidelines:

Multiple-choice Answer key

1	C
2	A
3	A
4	B
5	D
6	A
7	B
8	A
9	A
10	D

1	$\underline{u} = 3\underline{i} - \underline{j} - 7\underline{k}$ and $\underline{v} = 2\underline{i} + 3\underline{j} + \underline{k}$ $\underline{u} \bullet \underline{v} = 3(2) + (-1)(3) + (-7)(1)$ $\underline{u} \bullet \underline{v} = -4$ (C)	Correct answer:
2.	$\frac{1}{1-z} = \frac{1}{1-(3-4i)}$ $= \frac{1}{-2+4i} \times \frac{-2-4i}{-2-4i}$ $= \frac{-2-4i}{20}$ $= \frac{-1-2i}{10}$ (A)	Correct answer:
3.	Centre (2, 2, 3) and touches the $x-y$ plane, i.e. $z = 0$ i.e. radius is 3 units $(x-2)^2 + (y-2)^2 + (z-3)^2 = 9$ (A)	Correct answer:
4.	$\frac{16x-43}{(x-3)(x^2-x-6)} = \frac{16x-43}{(x-3)(x-3)(x+2)}$ $= \frac{16x-43}{(x-3)^2(x+2)}$ $= \frac{A}{x-3} + \frac{B}{(x-3)^2} + \frac{C}{x+2}$ $= \frac{A(x-3)(x+2) + B(x+2) + C(x-3)^2}{(x-3)^2(x+2)}$ $16x-43 = A(x-3)(x+2) + B(x+2) + C(x-3)^2$ When $x = 3$, $5 = 5B$ $\therefore B = 1$ $x = -2$, $-75 = 25C$ $\therefore C = -3$ $x = 0$, $-43 = -6A + 2 - 27$ $-18 = -6A$ $\therefore A = 3$ $\therefore \frac{3}{x-3} + \frac{1}{(x-3)^2} - \frac{3}{x+2}$ (B)	Correct answer:

5.	 $z-1 =1$ $z-\bar{z} \geq 1$ $z-\bar{z} = 2i < 2$ $i < 1$ $\{z : z-\bar{z} < 2\} \cap \{z : z-1 \geq 1\}$ (D)	Correct answer:
6.	$\int (\ln x)^2 dx = uv - \int u'v dx$ $u = (\ln x)^2$ $v' = 1$ $u' = \frac{2 \ln x}{x}$ $v = x$ $= x(\ln x)^2 - 2 \int \ln x dx$ (A)	Correct answer:
7.	 Let $\arg(z_1) = \theta$ Angle between z_1 and line $l = \alpha - \theta$ $\therefore \arg(z_1) + \arg(z_2) = \theta + (\alpha + \alpha - \theta)$ $= 2\alpha$ (B)	Correct answer:
8.	$\forall P = \text{both } m \text{ and } n \text{ are divisible by } d$ $Q = m - n \text{ is divisible by } d$ Converse: If $Q \Rightarrow P$ (A)	Correct answer:

9.

Key words	Negation
$\forall x \in R$, is even	$\exists x \in R$, is not even

$\exists x \in R$, if $f(x)$ is not even, then $f''(x)$ is also not even **(A)**

Correct answer:

10.

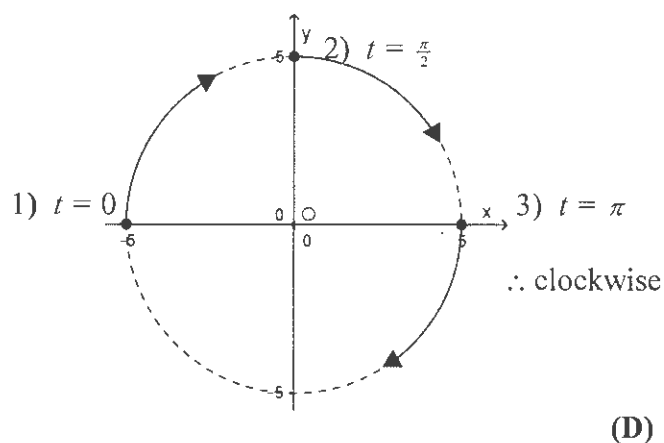
$$x = -5 \cos t$$

$$y = 5 \sin t$$

$$z = t$$

i.e. spiral up z - axis as t increases.

Method 1: viewing from the $x - y$ plane



Method 2: Table of values

	$t = 0$	$t = \frac{\pi}{2}$	$t = \pi$
$x = -5 \cos t$	$x = -5$	$x = 0$	$x = 5$
$y = 5 \sin t$	$y = 0$	$y = 5$	$y = 0$
$z = t$	$z = 0$	$z = \frac{\pi}{2}$	$z = \pi$

$\therefore$ turning clockwise around the z axis.

Correct answer:

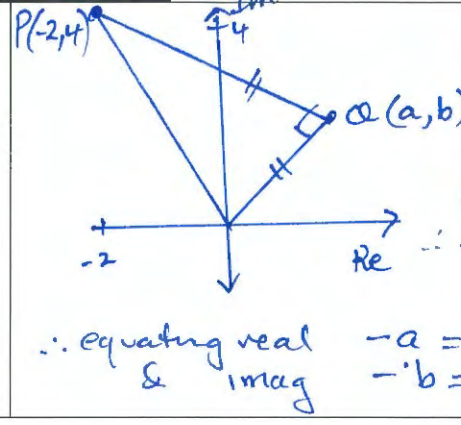
QUESTION 11 (a)

(i)	$(\sqrt{a} - \sqrt{b})^2 \geq 0$ $a - 2\sqrt{ab} + b \geq 0$ $a + b \geq 2\sqrt{ab}$ $a > 0, b > 0$ equality when $a = b$	Mostly well done All but 1 got it correct (2)
(ii).	$\sec^2 x = \tan^2 x + 1$ $\geq 2\sqrt{\tan^2 x \cdot 1} \quad (\text{using (i)})$ $= 2 \tan x $ $= 2\tan x \quad \text{as } \tan^2 x \geq 0$	Mostly well done All but 2 got it correct (1)

QUESTION 11 (b)

(i)	$z = \sqrt{2} - i\sqrt{2}$ $ z = \sqrt{2+2} = 2$ $\arg z = \tan^{-1}\left(\frac{-\sqrt{2}}{\sqrt{2}}\right)$ $= \tan^{-1}(-1)$ $= -\frac{\pi}{4}$ $\therefore z = 2 \operatorname{cis}\left(-\frac{\pi}{4}\right)$ $= 2\left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4}\right)$	Well done 1 incorrect 2
(ii)	$z^{24} = 2^{24} \operatorname{cis}\left(-\frac{\pi}{4}\right)^{24}$ $= 2^{24} \operatorname{cis}(-6\pi)$ $= 2^{24} (\cos 6\pi - i \sin 6\pi)$ $= 2^{24} (1 - 0)$ $= 2^{24} \quad (16777216)$	Well done 2 incorrect 1

QUESTION 11 (c)

 $\vec{OP} = -2 + 4i$ $\vec{OQ} = a + bi$ $\vec{QP} = \vec{OP} - \vec{OQ} = (-2 + 4i) - (a + bi)$ $\vec{QR} = i \vec{QP}$ $\therefore -(a + bi) = i[-2 + 4i] - (a + bi)$ $= (b - 4) + (-2 - a)i$ $\therefore \text{equating real \& imag } \begin{cases} -a = b - 4 \\ -b = -2 - a \end{cases} \text{ gives } a = 1, b = 3$ $\therefore \vec{OQ} = 1 + 3i$	A lot of students had $OP \perp OQ$ when it is $PQ \perp OQ$ Not many got this correct
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QUESTION 11 (d)

(i) $\int \frac{e^x}{1+e^{2x}} dx = \int \frac{1}{1+u^2} du \quad \text{let } u = e^x$ $du = e^x dx$ $= \tan^{-1} u + c$ $= \tan^{-1} e^x + c$	generally well done All but 2 got it correct
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QUESTION 11 (d)

(ii) $\int \sin^2 x \cos^3 x dx = \int \sin^2 x \cos^2 x \cos x dx$ $= \int \sin^2 x (1 - \sin^2 x) \cos x dx$ $= \int \cos x \sin^2 x - \cos x \sin^4 x dx$ $= \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + c$	generally well done All but 3 got it correct
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QUESTION 11 (d)

(iii) $\int \frac{2x+1}{x^2+2x+2} dx = \int \frac{2x+2-1}{x^2+2x+2} dx$ $= \int \frac{2x+2}{x^2+2x+2} dx - \int \frac{1}{x^2+2x+2} dx$ $= \ln(x^2+2x+2) - \int \frac{1}{(x+1)^2+1} dx$ $= \ln(x^2+2x+2) - \tan^{-1}(x+1) + c$	generally well done all but 3 got it correct
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QUESTION 12 (a)

(i)	Let $P(z) = 2z^3 - 5z^2 + 6z - 2$ $P(1-i) = 2(1-i)^3 - 5(1-i)^2 + 6(1-i) - 2$ $P(1-i) = 2(1-3i+3i^2-i^3) - 5(1-2i+i^2) + 6-6i-2$ $= 2(1-3i-3+i) - 5(1-2i-1) + 4-6i$ $= 2(-2-2i) - 5(-2i) + 4-6i$ $= -4-4i+10i+4-6i$ $= 0$ (1 mark) Hence $1-i$ is a root of $P(z) = 2z^3 - 5z^2 + 6z - 2$. <i>coefficients of $P(z)$ are real,</i>	DNK: 1/1 Expansion error $(-i)^3 = 1$ Poor working out: 1 No conclusion: 1
(ii)	Since $1-i$ is a root, then its conjugate $1+i$ is also a root <i>1</i> i.e. $(1-i) + (1+i) = 2$ $(1-i)(1+i) = 2$ $\therefore [z - (1-i)][z - (1+i)] = z^2 - 2z + 2$ is a factor $\begin{array}{r} 2z-1 \\ z^2-2z+2 \overline{) 2z^3-5z^2+6z-2} \\ \underline{-(2z^3-4z^2+4z)} \\ -z^2+2z-2 \\ \underline{-(-z^2+2z-2)} \\ 0 \end{array}$ Hence the other roots are $1+i$ and $\frac{1}{2} + 0i$ over $\mathbb{C}$.	ϕ 1 AE: 1 Had $\alpha + \beta + \gamma = -\frac{d}{a}$ 1/1 did not explain why $1-i$ is Had $1-i$ as the other roots: Did not answer the question for the other roots: Factored $P(z)$ as answer: 1/1 DNK: 1

QUESTION 12 (b)

(i)	'If $5^x = 2$, $x \in R$, then x is irrational.' where $P = '5^x = 2, x \in R'$ and $Q = x$ is irrational The converse statement (i.e. $Q \Rightarrow P$) is 'If x is irrational, $x \in R$, then $5^x = 2$.'	Wrong: forgot '$x \in R$':
(ii)	Using contradiction, 'If $5^x = 2$, $x \in R$, then x is rational.' If $5^x = 2$, $x \in R$, where x is rational i.e. $x = \frac{p}{q}$ where $p, q \in Z^+$ and $\frac{p}{q}$ in simplest form. $5^{\frac{p}{q}} = 2$ $\left(5^{\frac{p}{q}}\right)^q = (2)^q$ $5^p = 2^q$ Since LHS = 5^p $= 5 \times 5 \times 5 \times \dots \times 5$ where there is no factor that is divisible by 2. <i>or $5 \times 5 \times 5 \times \dots$ is odd.</i> and RHS = 2^q $= 2 \times 2 \times 2 \times \dots \times 2$ where there is no factor that is divisible by 5. <i>or $2 \times 2 \times 2 \times \dots$ is even.</i> Hence the proposal of $x = \frac{p}{q}$ is rational, is a contradiction therefore the original statement is true. <i>OR</i>	<i>Did not state contradiction: </i> <i>Did not explain fully: </i> <i>φ 1</i> <i>DNK: 1</i> <i>Proven by contrapositive then said statement is a contradiction: 1</i> <i>Conceptual problem with definition: </i> <i>Proven by contradiction not concluded by converse: 1</i> <i>Got contrapositive statement wrong: 1</i>

By Contrapositive "If x is rational, then $5^x \neq 2$, $x \in R$ ".

No conclusion!

QUESTION 12 (c)

Let $\int x \sin^{-1} x \, dx$

$= uv - \int u'v \, dx$ where $u = \sin^{-1} x$ and $v' = x$

$u' = \frac{1}{\sqrt{1-x^2}} \rightarrow v = \frac{x^2}{2}$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \frac{x^2}{\sqrt{1-x^2}} \, dx$ Let $x = \sin \theta$

$\frac{dx}{d\theta} = \cos \theta$

$dx = \cos \theta \, d\theta$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \frac{\sin^2 \theta}{\sqrt{1-\sin^2 \theta}} \cos \theta \, d\theta$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \frac{\sin^2 \theta}{\sqrt{\cos^2 \theta}} \cos \theta \, d\theta$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \frac{\sin^2 \theta}{|\cos \theta|} \cos \theta \, d\theta$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \sin^2 \theta \, d\theta$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \int \frac{1}{2} (1 - \cos 2\theta) \, d\theta$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{4} \int 1 - \cos 2\theta \, d\theta$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{4} \left[\theta + \frac{\sin 2\theta}{2} \right] + C$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{4} \left[\theta + \frac{2 \sin \theta \cos \theta}{2} \right] + C$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{4} [\theta + \sin \theta \cos \theta] + C$

$= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{4} [\sin^{-1} x + x \sqrt{1-x^2}] + C$

DNK what to do with $\int \frac{x^2}{\sqrt{1-x^2}} \, dx : ||||$

Had $\cos \theta = \frac{1}{\sqrt{1-x^2}} : 1$

$\phi : 1$

DNK: $\int \sqrt{1-x^2} \, dx : ||||$

accepted: $\frac{\sin(2 \sin^{-1} x)}{8}$

DNK: $v' = \sin^{-1} x : 1$

DNK : ||

careless transcription : 1

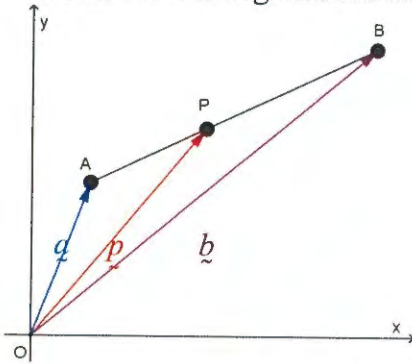
Had $\int u'v' \, dx : 1$

QUESTION 12 (c) continues

$$\begin{aligned}
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \sin^{-1} x + \frac{1}{2} \int \cos^2 \theta \, d\theta \\
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \sin^{-1} x + \frac{1}{2} \int \frac{1}{2} (1 + \cos 2\theta) \, d\theta \\
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \sin^{-1} x + \frac{1}{4} \int (1 + \cos 2\theta) \, d\theta \\
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \sin^{-1} x + \frac{1}{4} \left[\theta + \frac{\sin 2\theta}{2} \right] + C \\
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \sin^{-1} x + \frac{1}{4} \left[\sin^{-1} x + \frac{2 \sin \theta \cos \theta}{2} \right] + C \\
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \sin^{-1} x + \frac{1}{4} \left[\sin^{-1} x + \sin \theta \cos \theta \right] + C \\
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \sin^{-1} x + \frac{1}{4} \left[\sin^{-1} x + x \sqrt{1-x^2} \right] + C \\
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{2} \sin^{-1} x + \frac{1}{4} \sin^{-1} x + \frac{x \sqrt{1-x^2}}{4} + C \\
 &= \frac{x^2 \sin^{-1} x}{2} - \frac{1}{4} \sin^{-1} x + \frac{x \sqrt{1-x^2}}{4} + C
 \end{aligned}$$

QUESTION 12 (d)

- (i) Point P divides the line segment AB internally in the ratio of $m:n$.



Point P divides the line segment AB internally the ratio of $m:n$.

$$\overrightarrow{AB} = \overrightarrow{AP} + \overrightarrow{PB} \quad \text{where} \quad \overrightarrow{AP} : \overrightarrow{PB} = m : n$$

$$\frac{\overrightarrow{AP}}{\overrightarrow{PB}} = \frac{m}{n}$$

$$\therefore n \overrightarrow{AP} = m \overrightarrow{PB}$$

$$n(\overrightarrow{OP} - \overrightarrow{OA}) = m(\overrightarrow{OB} - \overrightarrow{OP})$$

$$n(\underline{p} - \underline{a}) = m(\underline{b} - \underline{p})$$

$$n\underline{p} - n\underline{a} = m\underline{b} - m\underline{p}$$

$$n\underline{p} + m\underline{p} = m\underline{b} + n\underline{a}$$

$$(n+m)\underline{p} = m\underline{b} + n\underline{a}$$

$$\therefore \underline{p} = \frac{m\underline{b} + n\underline{a}}{m+n}$$

DN ka: |||

Err in algebraic technique: !

$\phi: ||$

$$\text{Had } \overrightarrow{AP} = \frac{m}{n}(\underline{b} - \underline{a}) : 1$$

mostly did well using vectors properly.

Very division of interval but did not know how to relate to $\overrightarrow{OA}$ and $\overrightarrow{OB}$ as well as $\overrightarrow{OP}$: |||

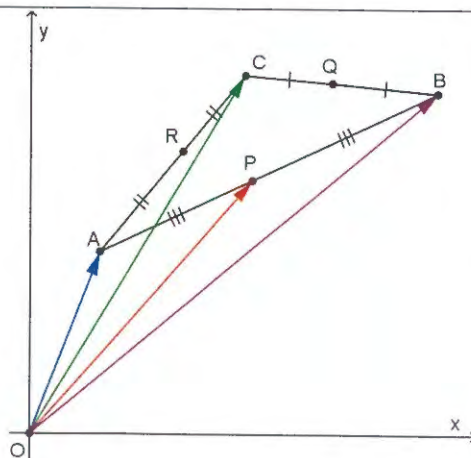
- (ii) Point P is the midpoint of AB , i.e. $m:n = 1:1$
 $\therefore m = n = 1$

$$\therefore \underline{p} = \frac{(1)\underline{b} + (1)\underline{a}}{(1) + (1)}$$

$$\therefore \underline{p} = \frac{\underline{b} + \underline{a}}{2}$$

Inadequate explanation: |||

$$\begin{aligned} & \underline{p} + \underline{q} + \underline{r} \\ &= \frac{\underline{a} + \underline{b}}{2} + \frac{\underline{a} + \underline{c}}{2} + \frac{\underline{b} + \underline{c}}{2} \\ &= \frac{2\underline{a} + 2\underline{b} + 2\underline{c}}{2} \\ &= \frac{2(\underline{a} + \underline{b} + \underline{c})}{2} \\ &\therefore \underline{p} + \underline{q} + \underline{r} = \underline{a} + \underline{b} + \underline{c} \end{aligned}$$



Mostly did well.

QUESTION 13 (a)

(i)	$A(4, -3, -3) \quad B(5, 2, 2)$ $\vec{AB} = \vec{OB} - \vec{OA}$ $= \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 4 \\ -3 \\ -3 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ 5 \\ 5 \end{pmatrix}$ $\therefore \text{Eqn of } l_1 \text{ (say } p): p = \begin{pmatrix} 4 \\ -3 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ 5 \end{pmatrix}$ * Note: there are others such as $p = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ 5 \end{pmatrix}$	Mostly well done
(ii)	$l_2: \frac{x-1}{1} = \frac{y}{-1} = \frac{z+3}{1} = \mu \text{ (say)}$ For l_1: $x = 4 + \lambda$ — ① $y = -3 + 5\lambda$ — ③ $z = -3 + 5\lambda$ — ⑤ For l_2: $x = 1 + k\mu$ — ② $y = -\mu$ — ④ $z = -3 + \mu$ — ⑥ from ⑤ and ⑥ $-3 + 5\lambda = -3 + \mu$ $\mu = 5\lambda$ * from ③ and ④ $-3 + 5\lambda = -\mu$ sub * $-3 + 5\lambda = -5\lambda$ $10\lambda = 3$ $\lambda = \frac{3}{10}$	Mostly well done except a few algebraic errors. Some need to learn method

then $\mu = \frac{3}{2}$ sub into ① & ②
 $4 + \lambda = 1 + k\mu$
 $4 + \frac{3}{10} = 1 + \frac{3}{2}k \Rightarrow k = \frac{11}{5}$

QUESTION 13 (b)

$\int_0^{\frac{\pi}{2}} \frac{dx}{3 - \cos x - 2 \sin x}$ $= \int_0^{\frac{\pi}{2}} \frac{1}{3 - \frac{1-t^2}{1+t^2} - \frac{4t}{1+t^2}} \cdot \frac{2dt}{1+t^2}$ $= 2 \int_0^1 \frac{dt}{3 + \frac{1-t^2}{1+t^2} - 1 + t^2 - 4t}$ $= 2 \int_0^1 \frac{dt}{4t^2 - 4t + 2}$ $= 2 \int_0^1 \frac{dt}{(2t-1)^2 + 1}$	let $t = \tan \frac{x}{2}$ $\frac{dt}{dx} = \frac{\sec^2 \frac{x}{2}}{2}$ $= \frac{1+t^2}{2}$ $x = \frac{\pi}{2} \quad t = 1$ $x = 0 \quad t = 0$ $= 2 \left[\tan^{-1} \frac{(2t-1)}{2} \right]_0^1$ $= \left[\tan^{-1}(2 \times 1 - 1) - \tan^{-1}(2 \times 0 - 1) \right]$ $= \tan^{-1}(1) - \tan^{-1}(-1)$ $= \frac{\pi}{4} - -\frac{\pi}{4} = \frac{\pi}{2}$	Generally well done to begin with. • Watch the bounds • constant divider was a problem for some
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QUESTION 13 (c)

(i)	$x^2 + y^2 + z^2 = 9 \text{ --- ①}$ $x^2 + (y-4)^2 + z^2 = 16 \text{ --- ②}$ $\textcircled{2} - \textcircled{1}$ $(y-4)^2 - y^2 = 7$ $y^2 - 8y + 16 - y^2 = 7$ $8y = 9$ $y = \frac{9}{8} \quad \therefore \text{they intersect on the plane } y = \frac{9}{8}$ --- ③	Really well done.
(ii)	sub ③ into ① $x^2 + \left(\frac{9}{8}\right)^2 + z^2 = 9$ $x^2 + \frac{81}{64} + z^2 = 9$ $x^2 + z^2 = \frac{495}{64}$ $\therefore$ Circle centre $(0, \frac{9}{8}, 0)$ radius $\frac{3\sqrt{55}}{8}u$	Mostly well done except for a couple of algebraic errors.

QUESTION 13 (d)

$$\text{RTP } \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}, \quad n \geq 2$$

① Prove for $n=2$

$$\begin{aligned} \text{LHS} &= \frac{1}{1^2} + \frac{1}{2^2} & \text{RHS} &= 2 - \frac{1}{2} \\ &= \frac{5}{4} & &= \frac{3}{2} \end{aligned}$$

as $\frac{5}{4} < \frac{3}{2}$ then true for $n=2$

② Assume true for $n=k$

$$\text{i.e. } \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} < 2 - \frac{1}{k}$$

③ Prove for $n=k+1$

$$\text{i.e. } \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1}$$

$$\text{LHS} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2}$$

$$< 2 - \frac{1}{k} + \frac{1}{(k+1)^2} \quad \text{using assumption}$$

$$= 2 + \frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} - \frac{1}{k+1}$$

$$< 2 - \frac{1}{k+1} \quad \text{as } \frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0 \quad \text{given}$$

< RHS

④ $\therefore$ if result is true for $n=k$, then it is true for $n=k+1$.

Since it is true for $n=2$, then it is true for $n=3$.

If it is true for $n=3$, then it is true for $n=4$ and so on.

$\therefore$ it is true for all integer $n \geq 2$.

Generally well done

• Some need to watch their step 1 as they were a little careless.

• Step 3 mostly well done

• Step 4 should be better even though marks are not generally awarded for it — however they can be deducted.

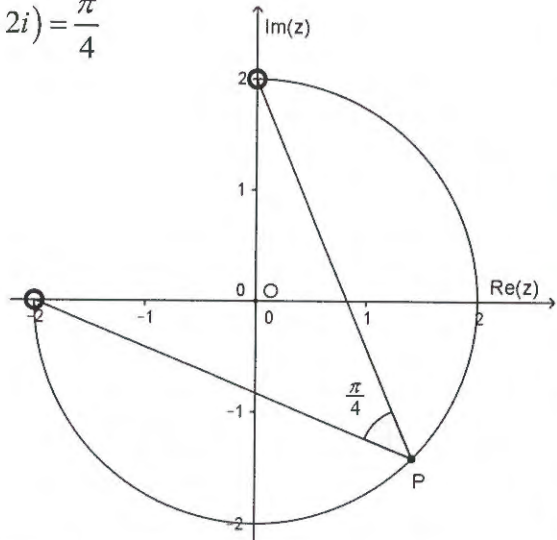
QUESTION 14 (a)

	'A quadrilateral is formed by joining any four points in a plane.' This statement is false when the four points are collinear, then a line is formed. OR This statement is false when the three points are collinear, then with the fourth point, a triangle is formed.	
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QUESTION 14 (b)

(i)	'If $\forall x \in R$, $x^2 - 6x + 5$ is even, then x is odd.' i.e. $P \Rightarrow Q$ <table border="1" data-bbox="202 504 1031 680"> <thead> <tr> <th>Given</th> <th>Negation</th> </tr> </thead> <tbody> <tr> <td>$\forall x \in R$</td> <td>$\exists x \in R$</td> </tr> <tr> <td>is even</td> <td>is not even</td> </tr> <tr> <td>is odd</td> <td>is not odd</td> </tr> </tbody> </table> $P =$ If $\forall x \in R$, $x^2 - 6x + 5$ is even $Q = x$ is odd Contrapositive: $\neg Q \Rightarrow \neg P$ $\therefore$ Contrapositive statement: If $\exists x \in R$, x is not odd, then $x^2 - 6x + 5$ is not even. OR If $\exists x \in R$, x is even, then $x^2 - 6x + 5$ is odd.	Given	Negation	$\forall x \in R$	$\exists x \in R$	is even	is not even	is odd	is not odd	4 used $\exists x \in R$
Given	Negation									
$\forall x \in R$	$\exists x \in R$									
is even	is not even									
is odd	is not odd									
(ii)	Let $x = 2n$, $n \in \mathbb{Z}$ (definition of even) $ \begin{aligned} x^2 - 6x + 5 &= (2n)^2 - 6(2n) + 5 \\ &= 4n^2 - 12n + 5 \\ &= 4n^2 - 12n + 4 + 1 \\ &= 2(2n^2 - 6n + 2) + 1 \\ &= 2m + 1, m \in \mathbb{Z} \text{ i.e. definition of odd.} \end{aligned} $ Since the original conjecture is false, the contrapositive is true. Hence the original conjecture is proven by contraposition.	many A far did not conclude properly to show that they understood how contraposition is related to original statement.								

QUESTION 14 (c)

(i)	$\arg(z+2) = \frac{\pi}{4} + \arg(z-2i)$ $\arg(z+2) - \arg(z-2i) = \frac{\pi}{4}$ 	Many students did not realise if $\angle APB = \frac{\pi}{4}$ then $\angle AOB = \frac{\pi}{2}$. A few forgot open circles.
(ii)	Centre is at (0, 0) Radius is 2 units.	

QUESTION 14 (d)

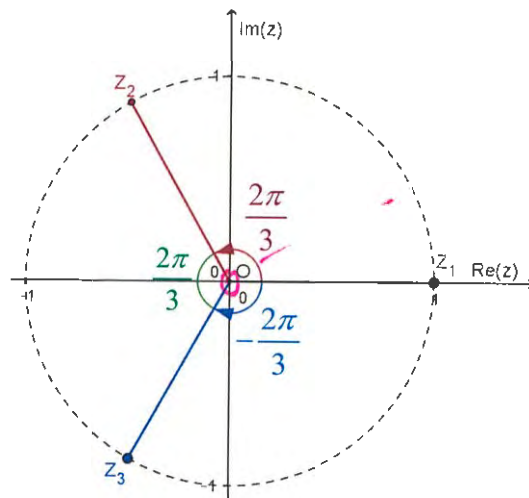
(i)	$z^3 - 1 = 0$ $z^3 = 1 \quad \text{where } z = \cos \theta + i \sin \theta$ $z^3 = \cos 3\theta + i \sin 3\theta \quad \text{by de Moivre's Theorem}$ $\cos 3\theta = 1$ $3\theta = 2k\pi + 0, \quad k = 0, 1, 2$ $\therefore \theta = \frac{2k\pi}{3}$ When $k = 0, \theta = 0, \therefore z_1 = 1$ $k = 1, \theta = \frac{2\pi}{3}, \therefore z_2 = \text{cis} \frac{2\pi}{3} = \omega$ $k = 2, \theta = \frac{4\pi}{3}, \therefore z_3 = \text{cis} \frac{4\pi}{3} = \text{cis} 2\left(\frac{2\pi}{3}\right) = \omega^2$ Hence ω^2 is a root of $z^3 - 1 = 0$.	Many approached from different methods. Some students still did not know how to do.
(ii)	ω is a complex root, hence $\omega^3 = 1$ $\omega^3 - 1 = 0$ $(\omega - 1)(1 + \omega + \omega^2) = 0 \quad \text{but } \omega \neq 1$ $\therefore 1 + \omega + \omega^2 = 0$	Mostly done well.

QUESTION 14 (d)

(iii) From (i) When $k = 0$, $\theta = 0$, $\therefore z_1 = e^{i0}$

$$k = 1, \theta = \frac{2\pi}{3}, \therefore z_2 = e^{\frac{2\pi}{3}i}$$

$$k = 2, \theta = \frac{4\pi}{3}, \therefore z_3 = e^{-\frac{2\pi}{3}i}$$



Open circle at origin

(iv) From (ii) $1 + \omega + \omega^2 = 0$

$$\text{i.e. } 1 + \omega = -\omega^2 \text{ or } \omega + \omega^2 = -1 \text{ or } 1 + \omega^2 = -\omega$$

$$\therefore (1 + \omega^2)^5 = (-\omega)^5$$

$$= -\omega^5$$

$$= -\omega^3 \times \omega^2 \text{ since } \omega^3 = 1$$

$$= -(1) \times \omega^2$$

$$= -\omega^2 \text{ (as required)}$$

*Expanded as binomial.
and got lost!*

QUESTION 15 (a)

(i)	given $a^2 + b^2 \geq 2ab$ then $x^2 + y^2 \geq 2xy$ — ① $x^2 + z^2 \geq 2xz$ — ② $y^2 + z^2 \geq 2yz$ — ③ so ① + ② + ③ $2x^2 + 2y^2 + 2z^2 \geq 2xy + 2xz + 2yz$ ①	generally well done
(ii)	So $x^2 + y^2 + z^2 \geq xy + xz + yz$ ✗ given $x + y + z = 1$ $(x + y + z)^2 = 1 = x^2 + y^2 + z^2 + 2(xy + xz + yz)$ $\therefore x^2 + y^2 + z^2 = 1 - 2(xy + xz + yz)$ $\therefore 1 - 2(xy + xz + yz) \geq xy + xz + yz$ (from (i)) $1 \geq 3(xy + xz + yz)$ $\therefore xy + xz + yz \leq \frac{1}{3}$ ②	generally well done

QUESTION 15 (b)

(i)	given $z = \cos \alpha + i \sin \alpha$ then $z^n = \cos n\alpha + i \sin n\alpha$ $z^{-n} = \cos(-n\alpha) + i \sin(-n\alpha)$ $= \cos n\alpha - i \sin n\alpha$ $\therefore z^n + z^{-n} = 2 \cos n\alpha$ ②	generally well done
(ii)	$(z + \frac{1}{z})^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$ $= (z^4 + \frac{1}{z^4}) + 4(z^2 + \frac{1}{z^2}) + 6$ $= 2 \cos 4\alpha + 4(2 \cos 2\alpha) + 6$ $= 2 \cos 4\alpha + 8 \cos 2\alpha + 6$ also $(z + \frac{1}{z})^4 = (2 \cos \alpha)^4$ $= 16 \cos^4 \alpha$ ③	generally well done
(iii)	$\therefore 16 \cos^4 \alpha = 2 \cos 4\alpha + 8 \cos 2\alpha + 6$ $\cos^4 \alpha = \frac{\cos 4\alpha}{8} + \frac{\cos 2\alpha}{2} + \frac{3}{8}$ $\therefore \int \cos^4 \alpha d\alpha = \int \frac{\cos 4\alpha}{8} + \frac{\cos 2\alpha}{2} + \frac{3}{8} d\alpha$ $= \frac{\sin 4\alpha}{32} + \frac{\sin 2\alpha}{4} + \frac{3\alpha}{8} + C$ ①	generally well done

QUESTION 15 (c)

Let $A = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ and P be a pt. on
 $r = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

Say $P = \begin{bmatrix} 1+2\lambda \\ -\lambda \\ 1+\lambda \end{bmatrix}$

$$\text{Now } \vec{AP} = \begin{bmatrix} 1+2\lambda \\ -\lambda \\ 1+\lambda \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2+2\lambda \\ -1-\lambda \\ -1+\lambda \end{bmatrix}$$

$|\vec{AP}|$ is minimum when $\vec{AP} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = 0$
 (right angles)

$$\therefore \begin{bmatrix} -2+2\lambda \\ -1-\lambda \\ -1+\lambda \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = 0 \Rightarrow -4+4\lambda-1-\lambda-1+\lambda=0$$

$$6\lambda-4=0 \Rightarrow \lambda=\frac{2}{3}$$

$$\text{then } P = \begin{bmatrix} 1+2 \cdot \frac{2}{3} \\ -\frac{2}{3} \\ 1-\frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{7}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

Poorly done question
 About 5 students
 did this correctly

QUESTION 15 (d)

$\therefore$ min distance is

$$\left| \begin{bmatrix} \frac{7}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \right| = \left| \begin{bmatrix} -\frac{2}{3} \\ -\frac{5}{3} \\ -\frac{1}{3} \end{bmatrix} \right|$$

$$= \sqrt{\left(-\frac{2}{3}\right)^2 + \left(-\frac{5}{3}\right)^2 + \left(-\frac{1}{3}\right)^2}$$

$$= \frac{\sqrt{30}}{3} \text{ UNITS}$$

(d) $a_1=2, a_2=3$ & $a_n=3a_{n-1}-2a_{n-2}$ for $n \geq 3$

Prove $a_n = 2^{n-1} + 1$

$$\text{for } n=1 \quad a_1 = 2^{1-1} + 1 = 1 + 1 = 2$$

$$n=2 \quad a_2 = 2^{2-1} + 1 = 2 + 1 = 3$$

$\therefore a_n = 2^{n-1} + 1$ is true for $n=1, 2$

$$\text{Assume true for } n=1 \quad a_1=2$$

$$n=2 \quad a_2=3$$

$$n=3 \quad a_3=2^2+1=5$$

$$n=k-1 \quad a_{k-1} = 2^{k-2} + 1$$

$$n=k \quad a_k = 2^{k-1} + 1$$

$$\text{Prove for } n=k+1, a_{k+1} = 2^{(k+1)-1} + 1$$

$$= 2^k + 1$$

$\leftarrow$ A lot of students
 did not do
 $n=1$ and $n=2$

The value of a_n
 is dependent on
 2 values preceding it

$\leftarrow$ Most did this correctly
 A few students did
 not show the
 sequence in the
 assumption.

Now $a_{k+1} = 3a_k - 2a_{k-1}$ by recurrence relation

$$= 3(2^{k-1} + 1) - 2(2^{k-2} + 1) \text{ by assumption}$$

$$= 3 \cdot 2^{k-1} + 3 - 2 \cdot 2^{k-2} - 2$$

$$= 3 \cdot 2^{k-1} - 2 \cdot 2^{k-2} + 1$$

$$= 3 \cdot 2^{k-1} - 2^{k-1} + 1$$

$$= 2 \cdot 2^{k-1} + 1$$

$$= 2^k + 1$$

Most did
this step
well.

1

which is true for $n = k+1$ if true for $n = 1 \leq n \leq k, k \geq 1$
where k is an integer

Since a_k is true for $n=1, n=2$ it is true for $n=3$

Since a_k is true for $n=2, n=3$ it is true for $n=4$

and so on

$\therefore a_n = 2^{n-1} + 1$ is true for all $n \geq 1$

This could
be a
lot better
for
nearly
everyone.

QUESTION 16 (a)

$$\int_0^{\frac{\pi}{4}} \sec^3 x \, dx = \int_0^{\frac{\pi}{4}} \sec^2 x \sec x \, dx$$

$$= [uv]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} u'v \, dx \text{ where } u = \sec x$$

$$u' = \sec x \tan x$$

$$v' = \sec^2 x$$

$$v = \tan x$$

$$= [\sec x \tan x]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \sec x \tan^2 x \, dx$$

$$= \left[\sec \frac{\pi}{4} \tan \frac{\pi}{4} - 0 \right] - \int_0^{\frac{\pi}{4}} \sec x (\sec^2 x - 1) \, dx$$

$$= [\sqrt{2}(1)] - \int_0^{\frac{\pi}{4}} \sec^3 x - \sec x \, dx$$

$$\text{Let } I = \int_0^{\frac{\pi}{4}} \sec^3 x \, dx$$

$$I = \sqrt{2} - \int_0^{\frac{\pi}{4}} \sec^3 x \, dx + \int_0^{\frac{\pi}{4}} \sec x \, dx$$

$$2I = \sqrt{2} + \int_0^{\frac{\pi}{4}} \sec x \, dx$$

$$2I = \sqrt{2} + \int_0^{\frac{\pi}{4}} \sec x \times \frac{\sec x + \tan x}{\sec x + \tan x} \, dx$$

$$2I = \sqrt{2} + \int_0^{\frac{\pi}{4}} \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx$$

$$2I = \sqrt{2} + [\ln |\sec x + \tan x|]_0^{\frac{\pi}{4}}$$

$$2I = \sqrt{2} + \left[\ln \left(\sec \frac{\pi}{4} + \tan \frac{\pi}{4} \right) - \ln(1+0) \right]$$

$$2I = \sqrt{2} + [\ln(\sqrt{2}+1) - \ln(\sec 0 + \tan 0)]$$

$$2I = \sqrt{2} + [\ln(\sqrt{2}+1) - \ln 1]$$

$$2I = \sqrt{2} + \ln(\sqrt{2}+1)$$

$$\therefore I = \frac{\sqrt{2}}{2} + \frac{1}{2} \ln(\sqrt{2}+1)$$

Mostly done well.

QUESTION 16 (b)

(i)	$l_1: r = (11\hat{i} + 2\hat{j} + 17\hat{k}) + \lambda(-2\hat{i} + \hat{j} - p\hat{k})$ $= \begin{bmatrix} 11 \\ 2 \\ 17 \end{bmatrix} + \lambda \begin{bmatrix} -2 \\ 1 \\ -p \end{bmatrix} \quad \text{with direction vector} \quad \begin{bmatrix} -2 \\ 1 \\ -p \end{bmatrix};$ $l_2: r = (-5\hat{i} + 11\hat{j} + q\hat{k}) + \mu(-3\hat{i} + 2\hat{j} + 2\hat{k})$ $= \begin{bmatrix} -5 \\ 11 \\ q \end{bmatrix} + \mu \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix} \quad \text{with direction vector} \quad \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix};$ For $l_1 \perp l_2$, $\begin{bmatrix} -2 \\ 1 \\ -p \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix} = 0$ $(-2)(-3) + (1)(2) - 2p = 0$ $6 + 2 - 2p = 0$ $8 - 2p = 0$ $-2p = -8$ $\therefore p = 4$	Many did not show that $\begin{bmatrix} -2 \\ 1 \\ -p \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix} = 0$.
(ii)	$r_1 = \begin{bmatrix} 11 \\ 2 \\ 17 \end{bmatrix} + \lambda \begin{bmatrix} -2 \\ 1 \\ -4 \end{bmatrix} \quad \text{and} \quad r_2 = \begin{bmatrix} -5 \\ 11 \\ q \end{bmatrix} + \mu \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix}$ Equating the like components: $11 - 2\lambda = -5 - 3\mu \quad \text{--- ①}$ $2 + \lambda = 11 + 2\mu \quad \text{--- ②}$ $17 - 4\lambda = q + 2\mu \quad \text{--- ③}$ From ② $\lambda = 9 + 2\mu$ --- ②' Sub ②' into ① $11 - 2(9 + 2\mu) = -5 - 3\mu$ $11 - 18 - 4\mu = -5 - 3\mu$ $\therefore \mu = -2 \quad \text{--- ④}$	Mostly well done.

QUESTION 16 (b)(ii) continues

(ii)	Sub ④ into ②' $\lambda = 9 + 2(-2)$ $\therefore \lambda = 5$ — ⑤ Sub ④ and ⑤ into ③ $17 - 4(5) = q + 2(-2)$ $-3 = q - 4$ $\therefore q = 1$	
(iii)	$r_1 = \begin{bmatrix} 11 \\ 2 \\ 17 \end{bmatrix} + 5 \begin{bmatrix} -2 \\ 1 \\ -4 \end{bmatrix}$ $r_1 = \begin{bmatrix} 11 - 10 \\ 2 + 5 \\ 17 - 20 \end{bmatrix}$ $r_1 = \begin{bmatrix} 1 \\ 7 \\ -3 \end{bmatrix}$ $\therefore$ The point of intersection is (1, 7, -3) Checking: $r_2 = \begin{bmatrix} -5 \\ 11 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix}$ $r_2 = \begin{bmatrix} -5 + 6 \\ 11 - 4 \\ 1 - 4 \end{bmatrix}$ $r_2 = \begin{bmatrix} 1 \\ 7 \\ -3 \end{bmatrix}$	<i>Did not answer as a "pair". Many did not answer as a pair.</i>

$$\begin{aligned}
 26 + 6\lambda &= \\
 6\lambda &= 26 - \frac{7}{13} \\
 &= \frac{331}{13}
 \end{aligned}$$

$\frac{169}{338} = \frac{1}{2}$
 $\frac{169}{338} = \frac{1}{2}$

QUESTION 16 (c)

(i)

$$\begin{aligned}\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} dx &= \left[\sin^{-1} x \right]_0^{\frac{1}{2}} \\ &= \sin^{-1} \frac{1}{2} - \sin^{-1} 0 \\ &= \frac{\pi}{6} - 0 \\ &= \frac{\pi}{6}\end{aligned}$$

all correct.

(ii)

For $n \geq 2$, since $0 \leq x \leq \frac{1}{2}$,

$$0 \leq x^n \leq \left(\frac{1}{2}\right)^n \quad \text{and} \quad 0 \leq x^2 \leq \left(\frac{1}{2}\right)^2$$

$$0 \leq \left(\frac{1}{2}\right)^n \leq \left(\frac{1}{2}\right)^2$$

$$\therefore 0 \leq x^n \leq x^2$$

many missed the point as $x \in \frac{1}{2}$

(iii)

For $n \geq 2$ and $0 \leq x \leq \frac{1}{2}$, $0 \leq x^n \leq x^2$

$$1 \geq 1 - x^n \geq 1 - x^2$$

$$1 \geq \sqrt{1 - x^n} \geq \sqrt{1 - x^2}$$

$$1 \leq \frac{1}{\sqrt{1 - x^n}} \leq \frac{1}{\sqrt{1 - x^2}}$$

$$\int_0^{\frac{1}{2}} 1 dx \leq \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1 - x^n}} dx \leq \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1 - x^2}} dx$$

$$\left[x \right]_0^{\frac{1}{2}} \leq \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1 - x^n}} dx \leq \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1 - x^2}} dx$$

$$\frac{1}{2} - 0 \leq \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1 - x^n}} dx \leq \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1 - x^2}} dx$$

$$\therefore \frac{1}{2} \leq \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1 - x^n}} dx \leq \frac{\pi}{6}$$

many did not start from $0 \leq x^n \leq x^2$ hence did not explain the inequality